

Analyzing Email Attachments

Summary

Let me be clear: this isn't a full malware analysis section. I'm only going to do some very basic triage-level malware analysis here. The point of this lesson is to show how I can triage suspicious files without doing a full-scale investigation.

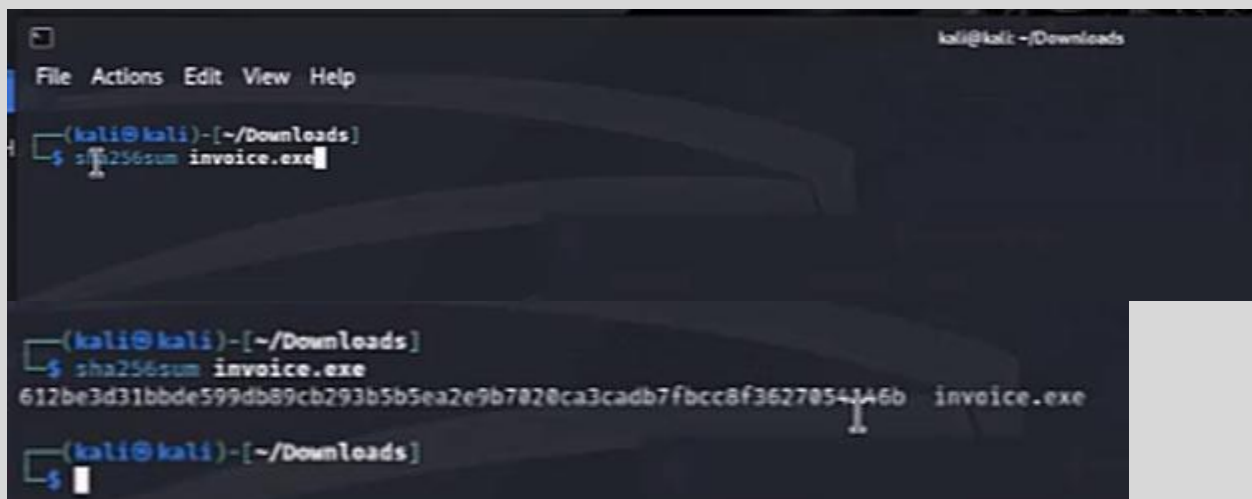
Lab Setup

Important Security Note: The file "invoice.exe" used here is a real malware sample. While I could just watch the video or submit it to sandboxes, I strongly recommend handling it carefully and only inside a Linux virtual machine. Since the malware is built for Windows, analyzing it in a Linux VM is much safer. I can download it from the link below (password: "infected"):

[Invoice.exe.zip](#)

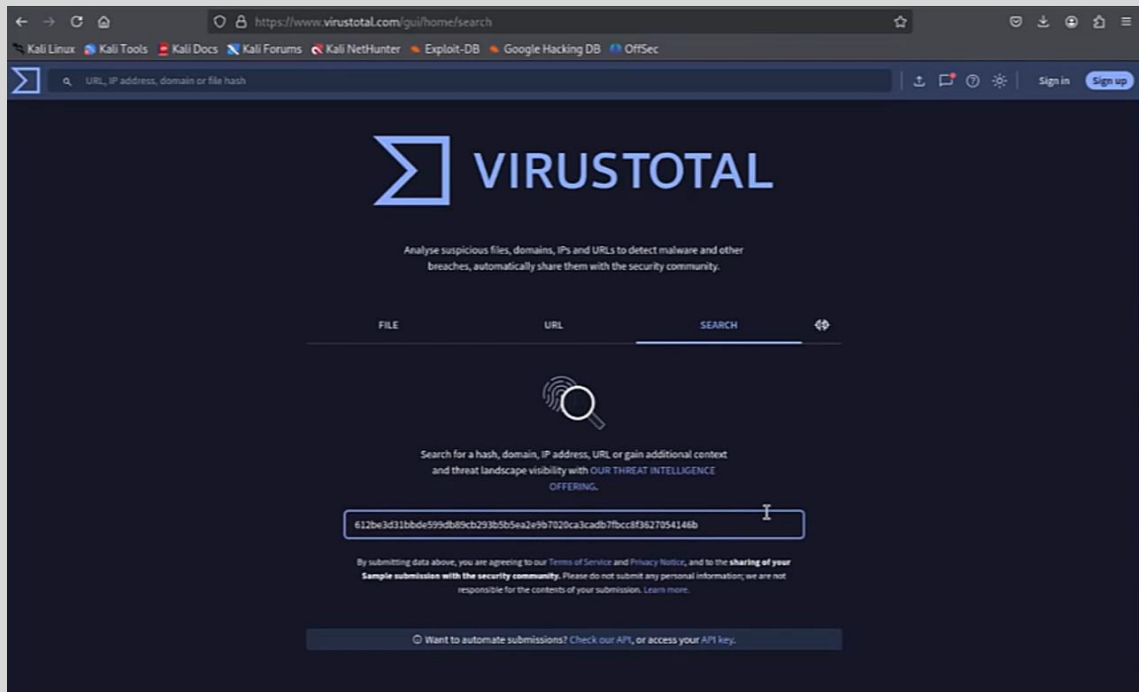
VirusTotal Analysis

First, I get the file hash by running this command in the directory where I saved the file:

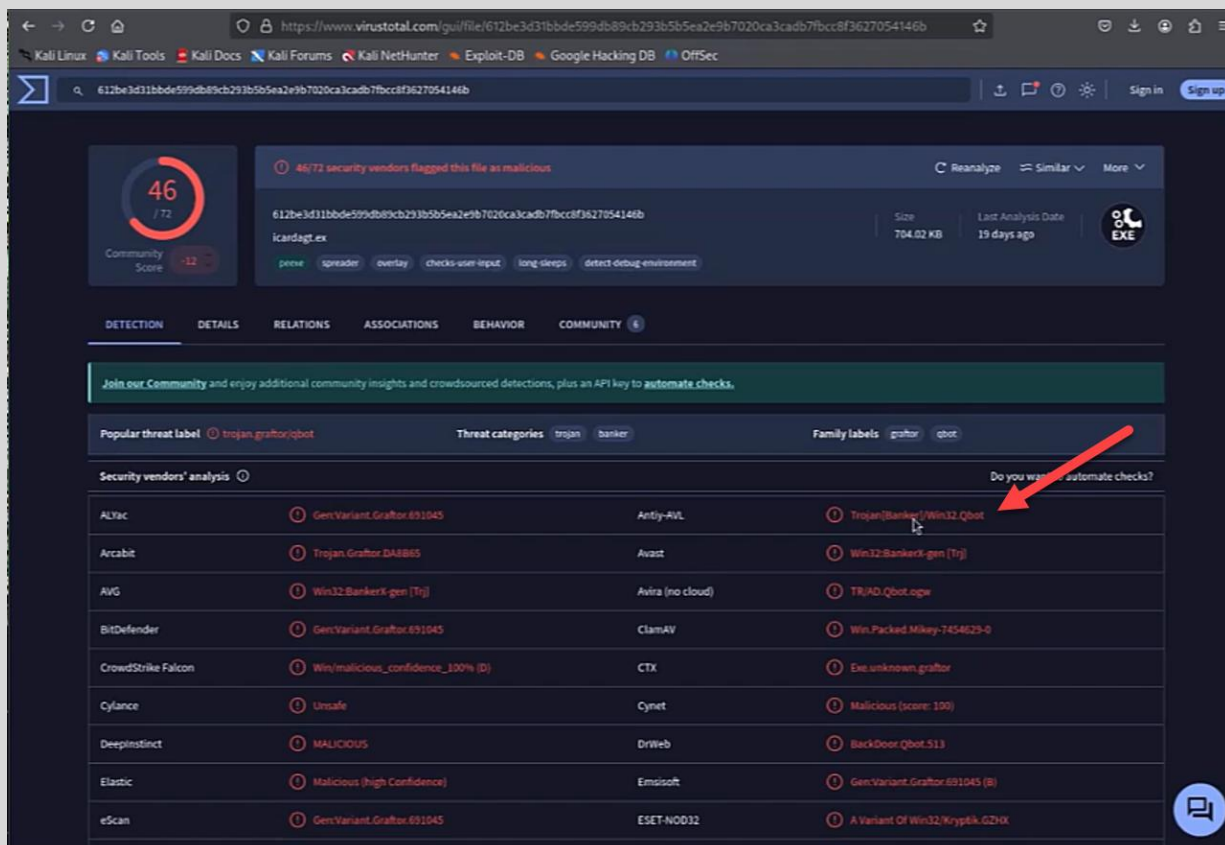


```
kali@kali: ~/Downloads
File Actions Edit View Help
(kali@kali)~[~/Downloads]
$ sha256sum invoice.exe
(kali@kali)~[~/Downloads]
$ sha256sum invoice.exe
612be3d31bbde599db89cb293b5b5ea2e9b7020ca3cadb7fbcc8f3627054346b invoice.exe
(kali@kali)~[~/Downloads]
$
```

I copy the hash and paste it into VirusTotal's search bar:



After hitting enter, I see that the file is flagged as malicious and identified as **Qbot** (aka Qakbot):



Online Sandbox Analysis

Using online sandboxes gives me a safe, fast, and cost-effective way to examine malicious software without risking my own systems. These platforms run samples in isolated, virtualized environments, so I don't have to worry about infections spreading. They generate automated reports that quickly show system changes, network activity, dropped files, and other behavioral indicators, which really speeds up my triage.

Key Benefits I Get from Online Sandboxes

1. Safe, Isolated Execution

- The malware runs in controlled environments, not on my own machines.
- I avoid any risk of infection to my personal or corporate network.

2. Rapid Automated Analysis

- I get comprehensive reports detailing exactly what the malware does.
- I can see file modifications, registry changes, network connections, and more in minutes.
- This helps me prioritize deeper investigations when needed.

3. Behavioral Monitoring

- The sandbox tracks file system changes, registry edits, process creation, and memory injections.
- I can spot hidden behaviors that static analysis might miss.
- I also see network communications and C2 traffic.

4. Multi-Environment Testing

- I can test in different OS versions (Windows, Linux, macOS) and configurations.
- Some sandboxes even simulate real-world user scenarios.

5. Network Traffic Capture

- I get PCAP files and DNS/HTTP logs to analyze network traffic.
- That helps me identify IOCs and track malicious domains or IP addresses.

6. Evasion Technique Detection

- Advanced sandboxes catch malware that tries to evade analysis:
 - Sandbox detection, anti-VM, anti-debugging, time-delayed execution.
 - Some use API hooking to catch stealthy behavior.

7. Threat Intelligence Integration

- Results are automatically correlated with threat intelligence feeds.
- I get malware family classification, signature matches, and links to similar samples.

8. Collaboration and Sharing

- Some platforms let me share reports with the community or submit samples to public databases.
- That fosters collaborative threat hunting and better security research.

9. No Infrastructure Requirements

- I don't need to set up and maintain a local malware lab (like FLARE-VM or Cuckoo).
- This is perfect for quick investigations or when local resources are tight.

10. Cost-Effectiveness

- Many services offer free tiers for basic analysis.
- Paid versions give me more features like longer execution times or deeper forensic reports.

Popular Online Sandboxes I Use:

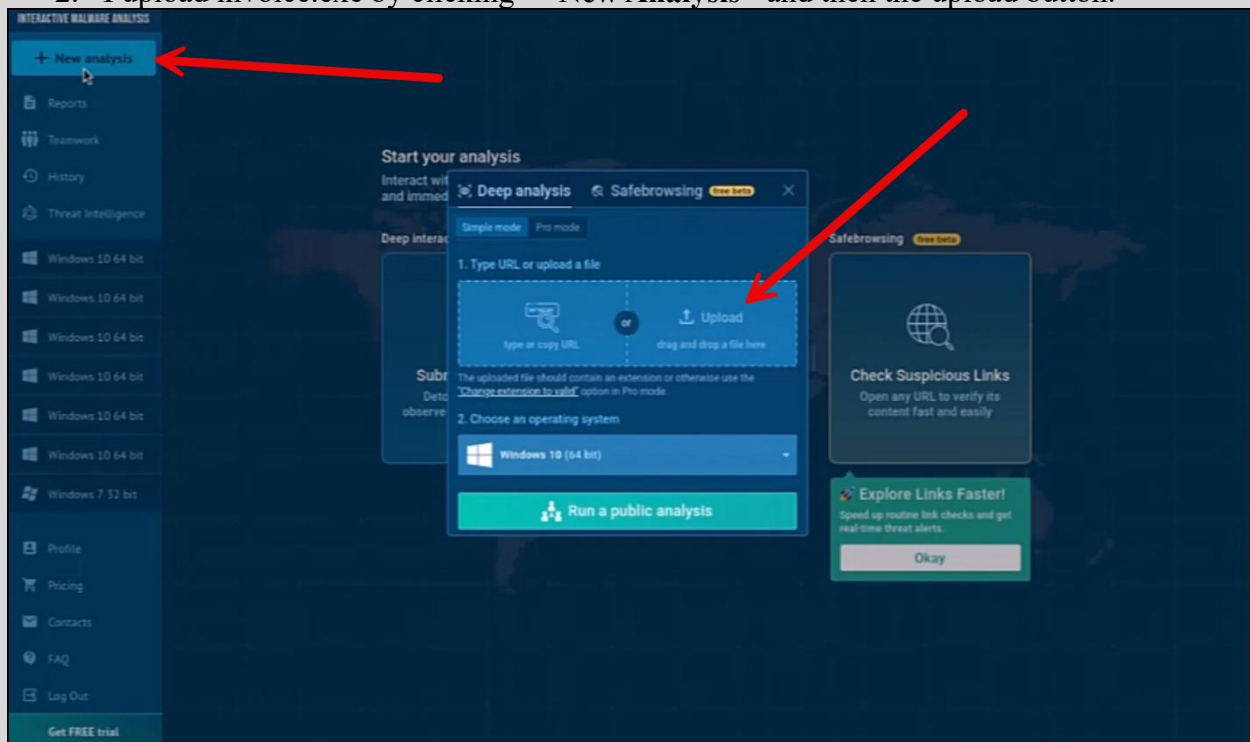
- **Any.Run** – Interactive, real-time analysis
- **Hybrid Analysis** – CrowdStrike-powered
- **VirusTotal Dynamic Analysis** – Google's behavioral sandbox
- **Joe Sandbox** – Advanced deep-dive
- **VxStream Sandbox** – Enterprise-grade automation
- **tri.age** – Recorded Future's high-speed platform

Any.Run Analysis

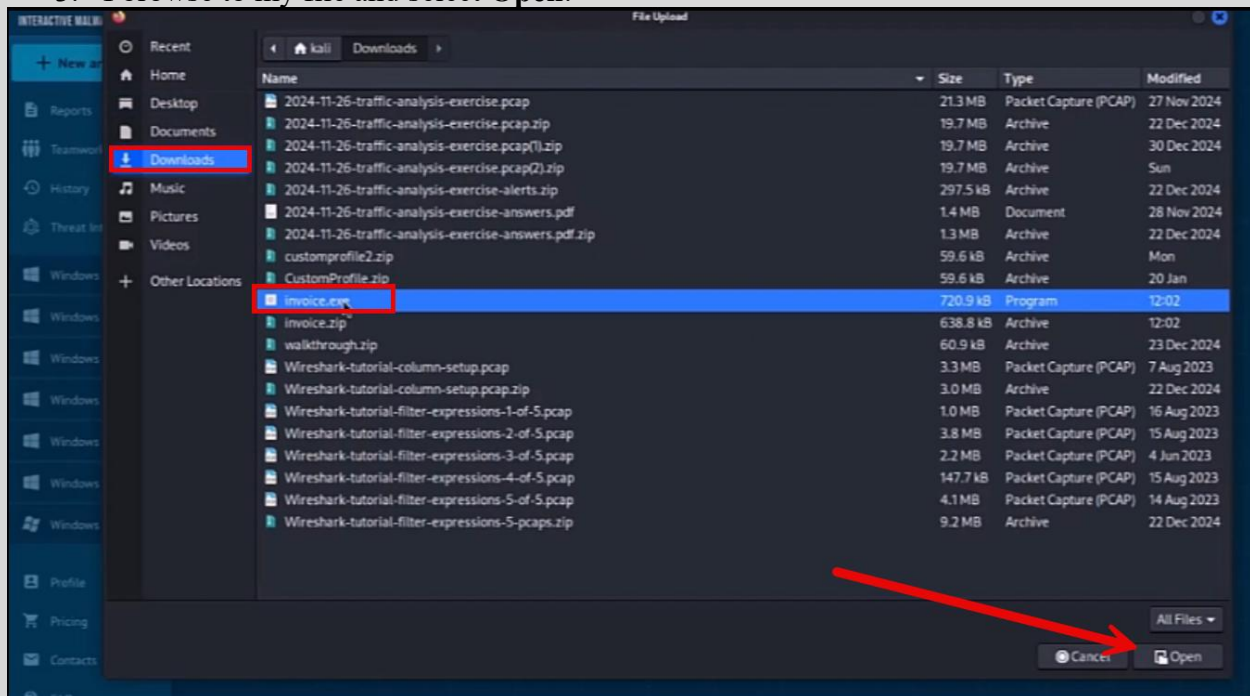
ANY.RUN is an interactive sandbox that lets me analyze suspicious files and URLs in a cloud-based environment. Unlike fully automated tools, ANY.RUN allows me to interact with the virtual machine in real time. I can manually execute malware, observe its behavior, and even click around the infected system. It gives me detailed process trees, network traffic logs, registry changes, and file system modifications, which is especially useful for evasive malware.

My Workflow:

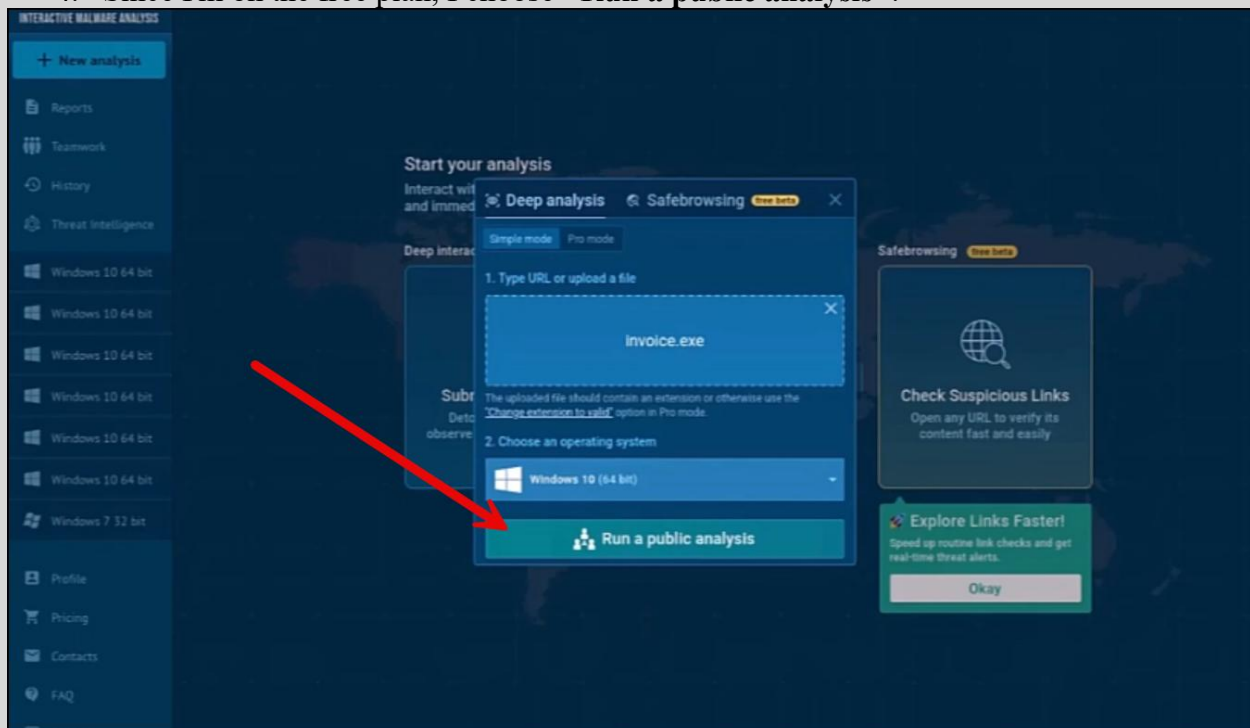
1. I navigate to <https://app.any.run/#register> and register (or sign in).
2. I upload invoice.exe by clicking "+New Analysis" and then the upload button.



3. I browse to my file and select **Open**.



4. Since I'm on the free plan, I choose **"Run a public analysis"**.



Results I Get:

The platform shows me multiple views:

Text Report: A detailed textual summary of findings.

Recorded IOCs

Download Sample

Full Report

Process Tree

Network Connections

MITRE ATT&CK

General Info

File name: invoice.exe

Full analysis: <https://app.any.run/tasks/259b131d-1294-4710-b000-00005971e2af>

Verdict: Malicious activity

Threats: Qbot Trojan

Qbot is a banking Trojan – a malware designed to collect banking information from victims. Qbot targets organizations mostly in the US. It is equipped with various sophisticated evasion and info-stealing functions, a worm-like functionality, and a strong persistence mechanism.

Analysis date: February 01, 2025 at 12:15:14

OS: Windows 10 Professional (build: 19045, 64 bit)

Tags: qbot, Trojan

Indicators:

MIME: application/vnd.microsoft.portable-executable

File info: PE32 executable (GUI) Intel 80386, for MS Windows, 7 sections

MD5: 40CD6343A9900F5C0484AA5EC45FAC5E

SHA1: 01DCAA199650529DE47C262D4310E1E7F80E848F

SHA256: 6128E3D31880E599089CB29385B5EA2E9B7026CA3C4D87F8C0F36276541468

SSDEEP: 24576:eq5A2to5et3/R7U5PZALHburLURJgmKcTa0OGKJtug+1Rpb8HSMz5A2to5et3/R7U5RALHburLURJgmK

Software environment set and analysis options

Behavior activities

MALICIOUS

QBOT mutex has been found

SUSPICIOUS

Starts a Microsoft application from unusual location

INFO

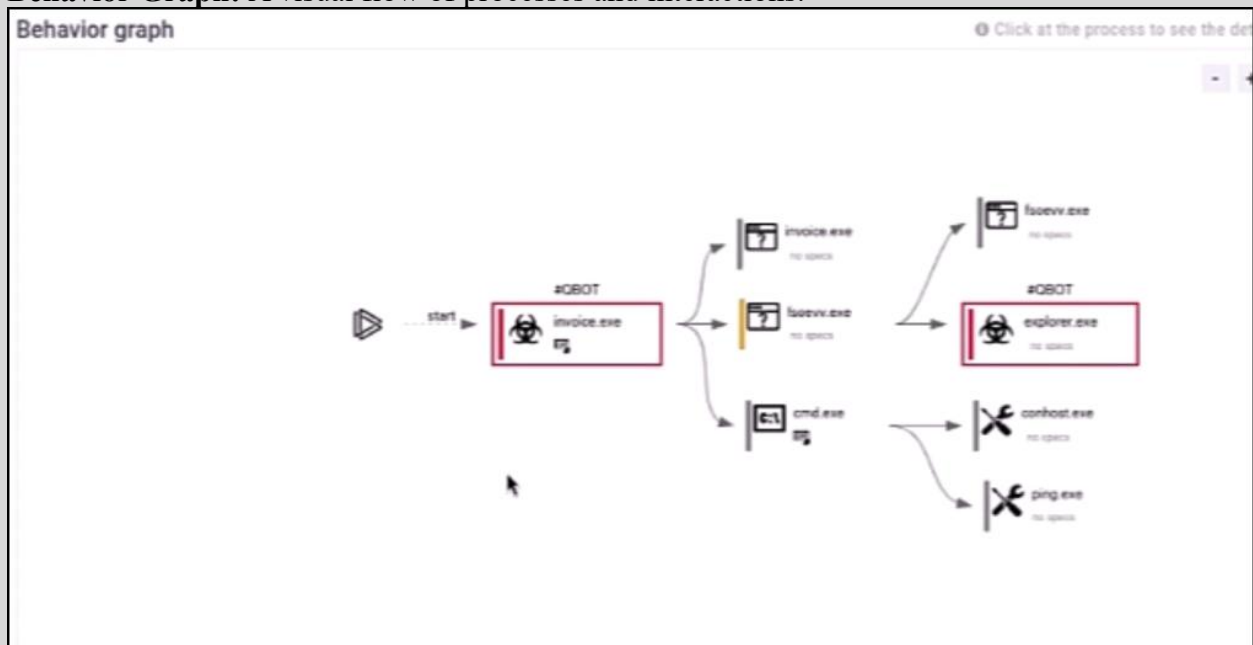
Checks supported languages

Indicators of Compromise (IOCs): All the IPs, domains, hashes, etc.

ATT&CK Framework Mapping: Which MITRE techniques the malware uses.

	Initial access	Execution	Persistence	Privilege escalation	Defense evasion	Credential access	Discovery	Lateral movement	Collection	C & C	Exfiltration	Impact
Home Reports Taskboard History TI OS & Sys ...		Command and Scripting Interpreter (1/6) Windows Command Shell 1			Masquerading (1/6) Rename System Utilities 2		Query Registry 1 System Information Discovery 8 System Location Discovery (2/1) 1					

Behavior Graph: A visual flow of processes and interactions.

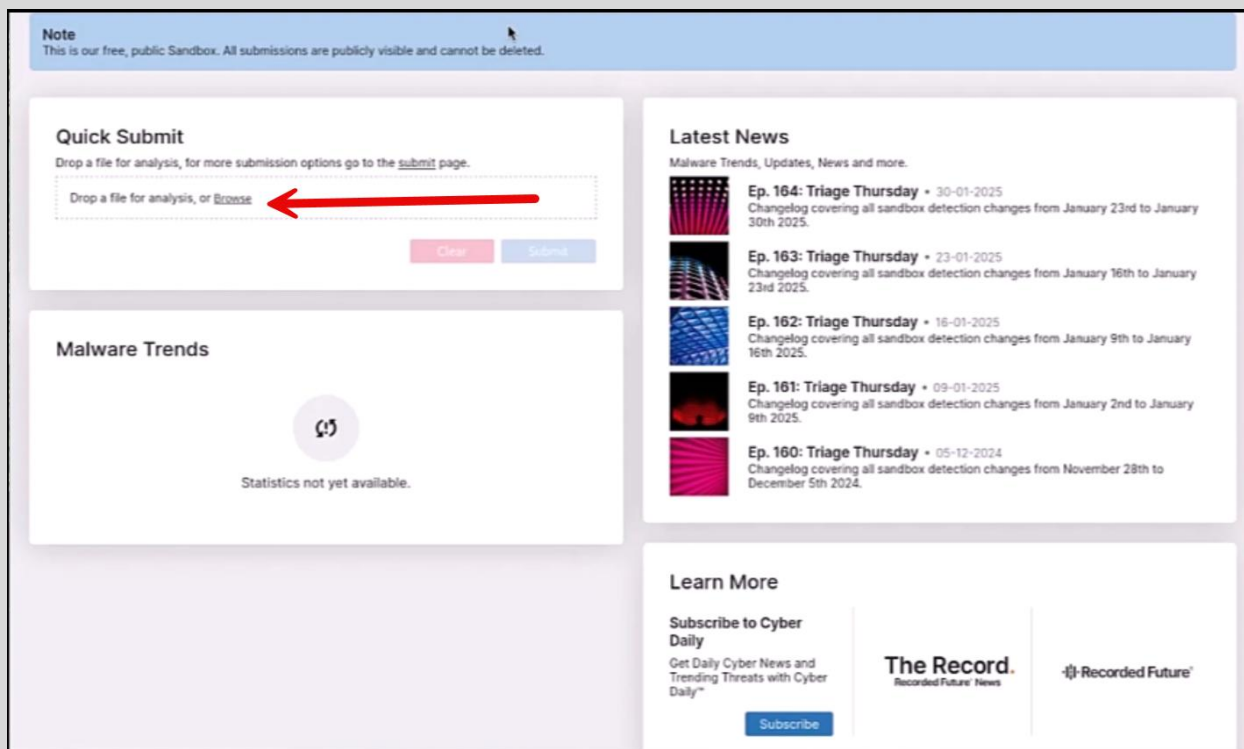


Triage Analysis

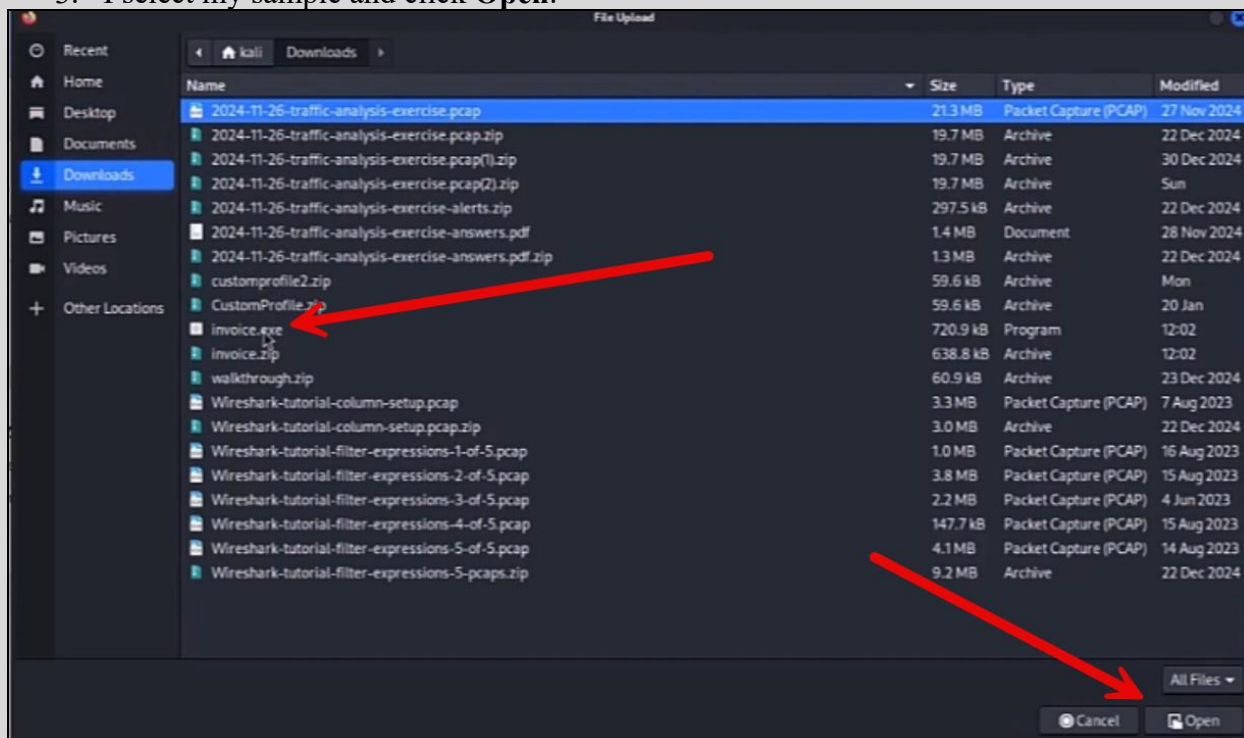
Recorded Future's Triage is another cloud-based sandbox I rely on for automated and interactive analysis of suspicious files, URLs, and scripts. It gives me rapid behavioral reports and extracts key IOCs like network connections, file changes, and registry modifications. It also integrates with Recorded Future's threat intelligence, enriching results with contextual data so I can quickly assess and respond.

My Workflow:

1. I go to <https://tria.ge/> and log in (or create an account).
2. In the **Quick Submit** section, I click **Browse**.



3. I select my sample and click **Open**.



4. Then I click **Submit**.

Note
This is our free, public Sandbox. All submissions are publicly visible and cannot be deleted.

File Code

Upload file

Upload a sample from your local machine for analysis, or paste a file on this window.

Name: invoice.exe
Size: 720.9kB
Type: application/x-msdownload

Optional Password
infected

The provided password is embedded in the static report.

Clear Submit

Submit from URL

Via URL submit you can retrieve files and URLs in various ways. These are the available options:

- **Analyze:** This is to directly analyze a URL that will be opened in the browser.
- **Fetch:** This fetches a file and executes the file in the sandbox.

URL/Fetch
http://example.com / http://example.com/malware.exe

Lock Analyze Fetch

API Access

As a Researcher you are eligible to use the REST API, enabling bulk submissions, automation of uploads and fetching analysis data, and full integration with other tools. For more information about the API check out our [documentation](#).

Apply for a Researcher license

Recorded Future © 2018-2025 Terms | Privacy

5. The initial score appears from a quick static analysis, I select **Analyze** to detonate the sample.

Submission Apply options for analysis Static static 3

Submission

Target
invoice.exe

Filesize
720.9kB

Submitted
01-02-2025 17:18

Password
N/A

Score
3/10

File tree

invoice.exe

Select All Deselect All

.EXE

Platforms

Automatic

Windows 7 x64 10-2004 x64 10-Itsc 2021 x64 11-21H2 x64

macOS 10.15 amd64

Android 10 x64 11 x64 13 x64 9 x86

Linux 12 amd64 12 mipsel 9 amd64 9 mips 9 mipsel 18.04 amd64 20.04 amd64 22.04 amd64 24.04 amd64

Languages

de-de en-us es-es fr-fr it-it ja-jp uk-ua

Internet Access

ON OFF Tor 20.04 40.4 DNS Disabled

Timeout

Analyze

Files selected: 1/32

Results:

The sample is flagged as malicious and identified as **Qakbot (QBot)**:

Resubmit

Download Sample

Download PCAP

Download PCAPNG

Feedback

Print to PDF

Analysis

max time kernel
129s

max time network
122s

platform
windows7_x64

resource
win7-20240903-en

Overview

Static

invoice.exe
windows7-x64

invoice.exe
windows10-2004-x64

Report

Analysis Logs

General

Target
invoice.exe

Size
704KB

MDS
40cd8343a9900f5c0484aa5ec45fac5e

SHA1
01d0aa199650529de47c262d4310e1e7f80eb4bf

SHA256
612be3d31bbde599db89cb293b5b5ea2e9b7020ca3cadb7fbcc8f3627054146b

SHA512
67ab7726e03800ab34a8862ca4fb231dfff5fde7106158115c7a871110d92d39f4d01d20864f662c32a615c8bd42dfc108fe4a8db9e6c13743fc2167d90d3c3f

SSDEEP
12288:a|4CKNIAJZ6Xs9W3FW198yMX8B|(fLxU:abKNIO6c43FS6yM20xU

Score

10¹⁰

QAKBOT

BANKER

PERSISTENCE

1575989975

DEFENSE_EVASION

STEALER

SPX44

DISCOVERY

TROJAN

I also get the malware's configuration file, which includes hardcoded IP addresses:

platform
windows7_x64

resource
win7-20240903-en

resource tags

ARCH:X64

ARCH:X86

IMAGE:WIN7-20240903-EN

LOCALE:EN-US

OS-WINDOWS7-X64

SYSTEM

submitted
01-02-2025 17:18

Sharing

Copy URL

Twitter

E-mail

12288:a|4CKNIAJZ6Xs9W3FW198yMX8B|(fLxU:abKNIO6c43FS6yM20xU

Malware Config

Extracted

Family
qakbot

Version
323.91

Botnet
spk44

Campaign
1575989975

C2

70.164.39.91:443

96.35.170.82:2222

96.37.137.42:443

75.70.218.193:443

73.226.220.56:443

104.152.16.45:995

24.184.6.58:2222

5.182.39.156:443

162.244.224.166:443

75.131.72.82:995

67.10.18.112:993

104.32.185.213:2222

181.126.80.118:443

75.131.72.82:443

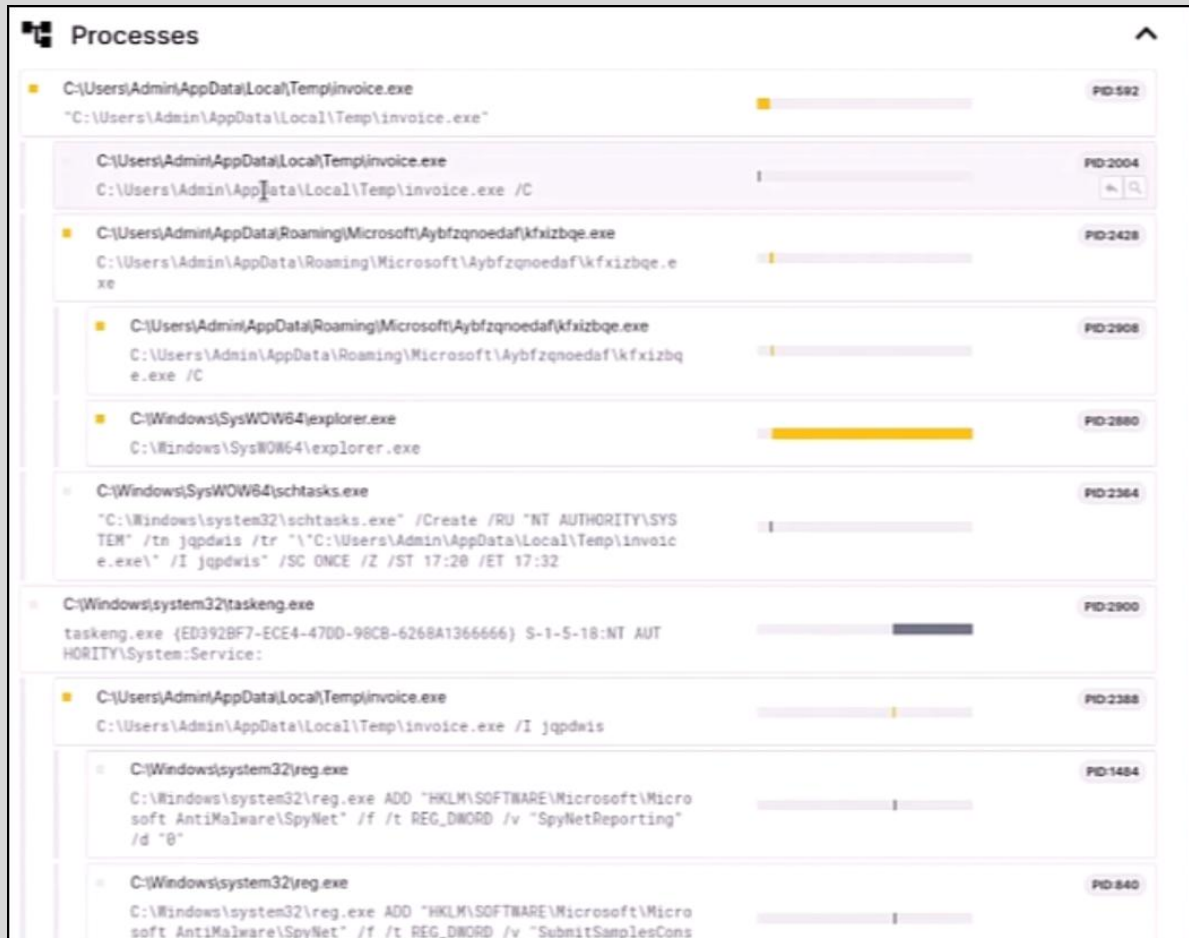
71.84.5.114:995

62.103.70.217:995

What I Learn About Qakbot's Behavior:

- It attempts to connect to the first IP in the list.
- If successful, it downloads the secondary payload and stops.
- If that IP fails, it moves to the next one, and so on.
- The malware is updated almost daily. developers know these IPs will eventually be blocked, so they include many fallback addresses.

Additionally, I see the processes it creates or modifies:



Conclusion:

Depending on my experience level, I could close the case at this point. It's Qakbot, I've blocked the IOCs, searched the network for any other deliveries, and I'm satisfied. But I'm not stopping here. I'll dive into dynamic analysis by executing the malware myself and analyzing its behavior firsthand.

Online sandboxes are fantastic for triage and quick assessments, but I know that nothing beats performing the dynamic analysis on my own. Automated reports give me a great starting point, but only by getting my hands dirty can I truly understand what the malware is up to.